

SESSION 7 MINI-ASSESSMENT REPORT

Design and Implementation of a Multi-Step Prompt Workflow for Resume Screening and Candidate Evaluation

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1. Introduction

Large language models can perform complex recruitment-support tasks more reliably when a large problem is divided into smaller prompt stages. Prompt chaining allows the output of one focused prompt to become the input of the next, improving consistency, traceability, and structure.

This mini-assessment designs a three-step prompt workflow for preliminary resume screening. The workflow reads an unstructured candidate profile, extracts job-relevant information, evaluates suitability against a sample Software Engineer job description, and finally converts the result into a machine-readable JSON object.

Workflow sequence:

Candidate Resume -> Information Extraction -> Suitability Assessment -> Structured JSON Output

Each stage has one clear responsibility: extraction, evaluation, and formatting.

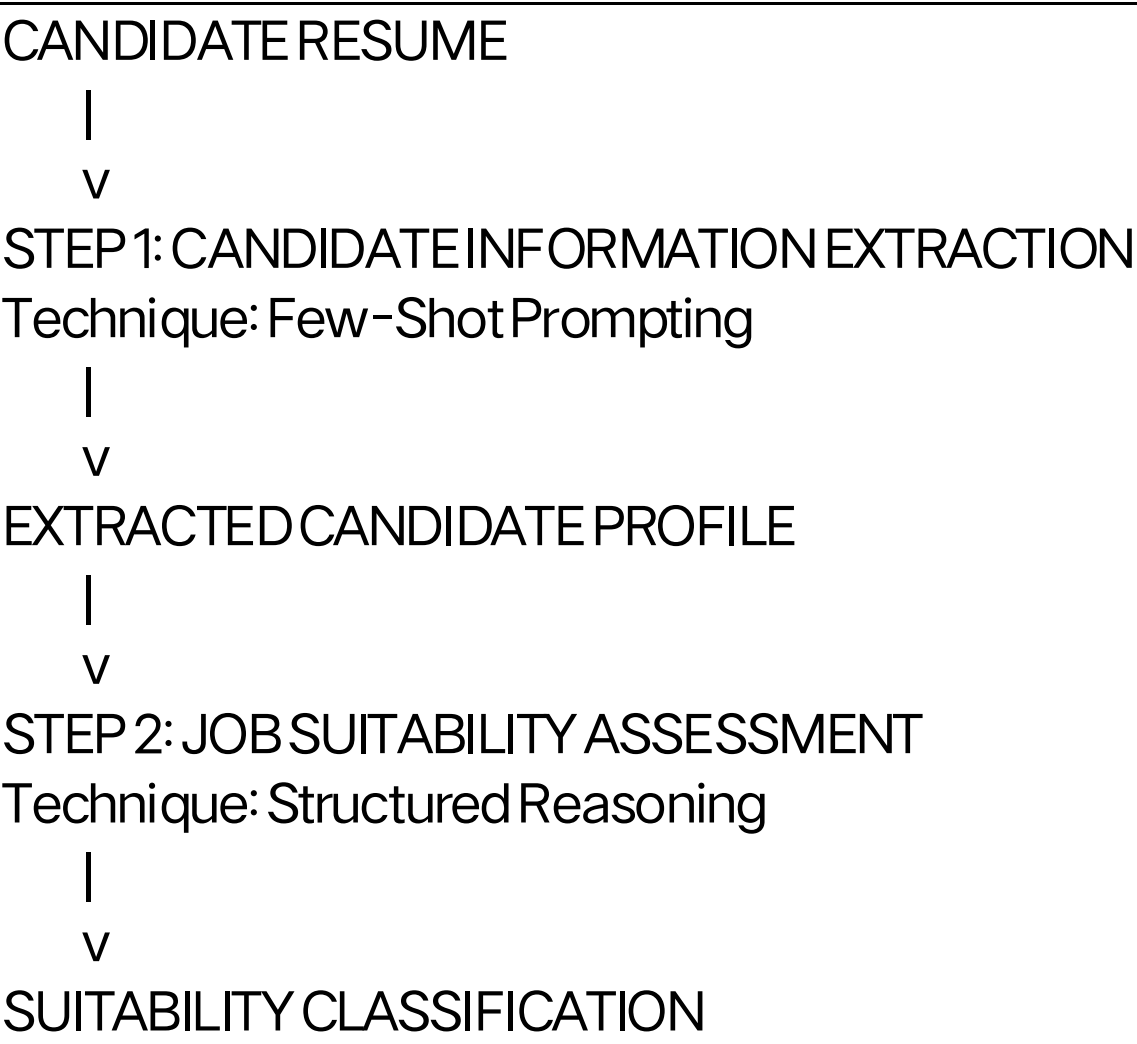
2. Objective

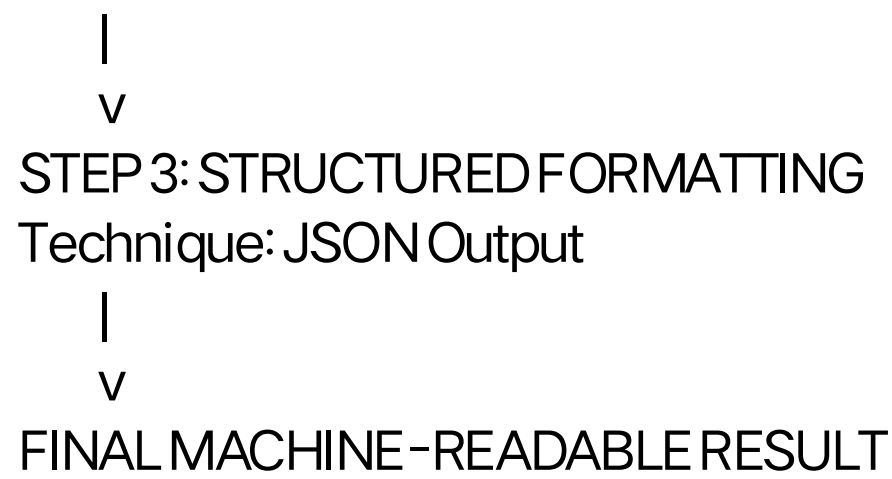
The objectives of this mini-assessment are to:

- 10. Extract important candidate information such as education, skills, projects and experience using few-shot prompting.
- 11. Compare the extracted information with a defined job requirement and classify candidate suitability.
- 12. Convert the results into a consistent JSON object that can be processed by recruitment software.
- 13. Demonstrate prompt chaining by passing the output of each stage to the next stage.

3. Workflow Design

The workflow is designed as a sequential three-stage pipeline:





Separating the workflow into stages ensures that the extraction prompt does not make hiring judgments, the evaluation prompt works only with extracted evidence, and the formatting prompt only structures the final result.

4. Methodology

The workflow can be implemented in any modern generative AI interface such as Google AI Studio. A fictional candidate profile and a fictional Software Engineer job description are used so the experiment is self-contained.

4.1 Step 1: Candidate Information Extraction Using Few-Shot Prompting

Purpose

The first stage extracts the candidate's key qualifications from unstructured resume text. Few-shot prompting is used to demonstrate the expected fields and output style.

Prompt Used

You are assisting with preliminary resume screening. Extract only information explicitly stated in the candidate profile. Use the examples as a guide.

Example 1:

Candidate: B.Tech CSE graduate. Skills: Java, SQL, Git. Built a library management system.

Output:

- Education: B.Tech CSE
- Skills: Java, SQL, Git
- Projects: Library Management System
- Experience: Not stated

Example 2:

Candidate: Final-year IT student with Python and Power BI skills. Completed a 2-month data analytics internship and created a sales dashboard.

Output:

- Education: Final-year IT student
- Skills: Python, Power BI
- Projects: Sales Dashboard
- Experience: 2-month Data Analytics Internship

Now analyze:

"Final-year B.Tech Computer Science (AI & ML) student. Skilled in Java, Python, SQL, HTML, CSS, JavaScript, Node.js and Git. Completed an AI/ML internship and a Full Stack Developer internship. Projects include OfficeVerse

AI, a To-Do web app and an E-commerce mini store. Familiar with machine learning and REST-based web applications."

Return:

- Education
- Skills
- Experience
- Projects
- Relevant knowledge

Output Generated

- Education: Final-year B.Tech Computer Science (AI & ML) student
- Skills: Java, Python, SQL, HTML, CSS, JavaScript, Node.js, Git
- Experience: AI/ML internship; Full Stack Developer internship
- Projects: OfficeVerse AI; To-Do web app; E-commerce mini store
- Relevant knowledge: Machine learning; REST-based web applications

Discussion

The first prompt converts unstructured resume text into a consistent candidate profile without yet deciding whether the candidate should be selected. This extracted profile becomes the input for Step 2.

4.2 Step 2: Job Suitability Assessment Using Structured Reasoning

Purpose

The second stage evaluates the extracted profile against explicit job requirements. The model is instructed to provide concise evidence-based reasons rather than unsupported assumptions.

Prompt Used

Evaluate the extracted candidate profile against this fictional Software Engineer job requirement.

Job requirements:

- Computer Science/IT or related degree
- Programming knowledge in Java or Python
- SQL/database fundamentals
- Web development fundamentals
- Git/version-control knowledge
- At least one relevant software project
- Internship exposure is preferred

Classification guide:

Highly Suitable: Meets nearly all core requirements and shows relevant project/internship exposure.

Suitable: Meets most core requirements but has some gaps.

Not Suitable: Misses several core requirements.

Candidate profile:
Education: Final-year B.Tech Computer Science (AI & ML)
Skills: Java, Python, SQL, HTML, CSS, JavaScript, Node.js, Git
Experience: AI/ML internship; Full Stack Developer internship
Projects: OfficeVerse AI; To-Do web app; E-commerce mini store
Relevant knowledge: Machine learning; REST-based web applications

Give 3-5 concise evidence-based reasons, list any gap, and finish exactly as:
Suitability: Highly Suitable / Suitable / Not Suitable

Output Generated

- 14. The candidate is pursuing a directly relevant Computer Science degree.
- 15. The profile includes both Java and Python as well as SQL.
- 16. HTML, CSS, JavaScript and Node.js demonstrate web-development exposure.
- 17. Git, multiple software projects and two relevant internships provide practical exposure.
- 18. A possible gap is that the profile does not state professional full-time software engineering experience.

Suitability: Highly Suitable

Discussion

The second stage uses only the structured evidence from Step 1 and the stated job criteria. This makes the decision easier to audit and prevents the evaluation stage from inventing missing resume details.

4.3 Step 3: Structured Output Using JSON

Purpose

The final stage converts the outputs from Steps 1 and 2 into a consistent machine-readable format suitable for a recruitment dashboard or applicant-tracking workflow.

Prompt Used

Convert the results from the previous two stages into one valid JSON object.
Return only JSON. Do not add markdown or explanations.
Use exactly these keys:
"candidate_profile", "suitability", "strengths", "gaps", "summary".

Candidate profile:
Education: Final-year B.Tech Computer Science (AI & ML)
Skills: Java, Python, SQL, HTML, CSS, JavaScript, Node.js, Git
Experience: AI/ML internship; Full Stack Developer internship
Projects: OfficeVerse AI; To-Do web app; E-commerce mini store
Relevant knowledge: Machine learning; REST-based web applications

Suitability: Highly Suitable

Gap: No full-time software engineering experience stated.

Final Output Generated

```
{
  "candidate_profile": {
    "education": "Final-year B.Tech Computer Science (AI & ML)",
    "skills": ["Java", "Python", "SQL", "HTML", "CSS", "JavaScript", "Node.js", "Git"],
    "experience": ["AI/ML internship", "Full Stack Developer internship"],
    "projects": ["OfficeVerse AI", "To-Do web app", "E-commerce mini store"]
  },
  "suitability": "Highly Suitable",
  "strengths": ["Relevant degree", "Programming and SQL", "Web development", "Git", "Projects and internships"],
  "gaps": ["No full-time software engineering experience stated"],
  "summary": "The candidate meets nearly all core requirements and demonstrates relevant project and internship exposure."
}
```

5. Results and Discussion

The three-stage workflow successfully transformed an unstructured candidate description into an extracted profile, an evidence-based suitability classification, and a structured JSON result.

Step	Task	Prompting Technique	Output
Step 1	Extract candidate details	Few-shot prompting	Structured candidate profile
Step 2	Assess job suitability	Structured reasoning	Suitability classification
Step 3	Format results	JSON output	Machine-readable JSON

The main advantage of this design is separation of responsibilities. Each intermediate output can be checked independently. The workflow is also reusable: changing the job description in Step 2 allows the same extracted profile to be evaluated for another role.

This workflow should be treated as decision support rather than an autonomous hiring system. Final recruitment decisions should include human review and should not infer protected or unstated personal characteristics.

6. Conclusion

This mini-assessment demonstrated a three-step prompt chaining workflow for resume screening and candidate evaluation. Few-shot prompting extracted candidate information, an evidence-based evaluation stage compared that information with explicit job requirements, and JSON formatting produced a consistent machine-readable result.

The completed sequence - Extract -> Evaluate -> Format - demonstrates how prompt chaining can make a complex GenAI task more modular, transparent and reusable. In a practical recruitment application, the workflow could support preliminary organization of applicant information while keeping final hiring decisions under human review.